

MARIANNA, FL, USA

WMO: 747760

Lat: 30.836N

Lon: 85.184W

Elev: 113

StdP: 14.64

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 03818

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.0	31.2	12.3	10.4	37.8	17.6	13.4	43.5	22.2	56.1	19.8	56.6	3.3	320	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	96.5	75.8	94.5	75.8	92.6	75.7	80.1	89.2	79.3	88.2	78.4	86.9	6.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.1	146.4	82.6	77.2	142.1	82.1	76.4	138.2	81.6	43.6	89.2	42.7	88.1	42.0	87.1	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	16.2	13.7	DB	22.2	99.1	3.5	3.1	19.7	101.3	17.6	103.1	15.7	104.8	13.1	107.1	
			WB	20.0	81.6	3.8	1.1	17.3	82.5	15.1	83.1	12.9	83.7	10.2	84.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.4	51.5	55.0	61.4	67.9	75.7	81.3	82.4	82.3	78.8	69.7	59.4	54.8	
	DBStd	13.10	10.04	8.80	8.10	6.30	5.27	3.43	2.84	2.99	4.30	7.30	8.24	9.13	
	HDD50	229	105	44	12	0	0	0	0	0	0	0	16	52	
	HDD65	1521	430	295	165	43	4	0	0	0	0	40	207	337	
	CDD50	6945	152	184	364	537	795	939	1003	1000	864	611	296	200	
	CDD65	2761	12	15	52	129	334	489	538	535	415	185	38	19	
	CDH74	26183	44	106	455	1183	3183	4873	5379	5277	3741	1592	268	82	
	CDH80	11078	0	13	78	313	1359	2299	2463	2379	1632	506	34	2	
Wind	WSAvg	5.4	6.3	6.2	6.2	6.0	5.6	5.0	4.4	4.4	5.0	5.0	5.1	5.7	
Precipitation	PrecAvg	53.2	4.1	4.6	5.3	4.6	3.3	5.3	5.7	6.0	3.9	2.4	3.3	4.5	
	PrecMax	69.8	8.1	12.9	8.8	11.4	8.6	8.9	11.8	14.5	10.4	6.9	7.2	11.2	
	PrecMin	36.4	1.1	0.8	1.0	1.3	1.1	1.2	2.9	2.2	0.9	0.5	0.8	0.4	
	PrecStd	11.0	1.6	3.0	2.6	2.6	2.0	2.2	2.2	3.6	2.2	1.8	1.8	2.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.4	82.1	85.4	89.5	96.0	99.8	97.6	98.4	96.7	91.2	83.5	79.7	
		MCWB	67.1	66.8	65.6	70.6	71.8	75.0	77.7	76.5	73.6	72.9	70.0	68.6	
	2%	DB	74.7	77.4	82.2	86.3	92.2	96.3	95.3	95.3	93.5	88.1	80.6	76.4	
		MCWB	65.0	64.6	65.8	68.7	71.4	75.3	76.8	77.4	74.3	71.9	68.4	68.0	
	5%	DB	71.4	73.1	79.3	83.6	90.0	93.7	93.4	93.1	91.1	85.6	77.1	73.1	
		MCWB	64.2	63.3	64.6	66.9	70.9	75.2	77.0	77.3	74.5	70.1	66.3	67.0	
	10%	DB	67.8	70.0	75.9	81.2	87.6	91.0	91.1	91.0	88.9	82.5	73.7	70.1	
		MCWB	62.0	61.9	62.7	66.0	70.4	75.0	76.8	77.1	74.2	69.2	65.4	65.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.5	70.6	71.5	74.3	76.4	80.2	81.2	81.4	79.4	78.0	75.5	73.2
			MCDB	73.7	75.2	76.7	80.7	85.1	89.5	89.9	91.4	87.3	84.2	78.2	75.7
2%		WB	67.9	68.6	69.8	72.6	75.2	78.9	80.1	80.2	78.2	76.2	72.6	70.4	
		MCDB	72.5	73.6	75.3	79.4	83.5	88.3	89.2	89.5	86.3	82.3	75.9	73.2	
5%		WB	65.7	66.3	68.1	71.0	74.2	78.1	79.2	79.4	77.4	74.6	70.1	68.3	
		MCDB	69.5	71.4	74.6	78.4	82.5	87.5	88.3	88.0	85.4	80.7	74.0	71.6	
10%		WB	63.1	64.0	66.0	69.3	73.2	77.2	78.3	78.4	76.3	72.9	67.2	66.0	
		MCDB	67.2	68.2	73.2	76.3	81.4	86.2	86.9	86.5	83.8	79.1	72.7	69.7	
Mean Daily Temperature Range	5% DB	MDBR	21.6	22.2	23.6	23.2	21.9	18.9	17.9	17.4	19.1	22.3	23.6	21.2	
		MCDBR	22.5	24.2	25.4	25.0	23.8	22.9	20.5	20.0	21.7	23.6	24.0	21.1	
		MCWBR	14.0	13.6	11.9	9.7	7.2	6.2	6.1	5.9	6.8	9.4	12.8	13.4	
	5% WB	MCDBR	18.5	19.7	20.4	19.5	19.1	18.6	18.1	17.6	17.3	17.6	18.4	17.7	
		MCWBR	13.6	14.1	12.1	10.1	7.0	6.5	6.5	6.2	6.3	8.5	11.9	13.3	
Clear-Sky Solar Irradiance	taub	0.327	0.339	0.360	0.387	0.440	0.481	0.507	0.500	0.436	0.379	0.351	0.337		
	taud	2.518	2.482	2.432	2.359	2.239	2.187	2.129	2.146	2.305	2.442	2.459	2.507		
	Ebn at Noon	287	293	293	287	271	259	252	252	265	275	275	277		
	Edh at Noon	28	31	35	39	45	47	49	48	40	32	29	27		
All-Sky Solar Radiation	RadAvg	975	1164	1551	1883	2052	1910	1850	1719	1580	1396	1097	856		
	RadStd	65	143	121	151	141	148	121	125	115	139	89	86		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+293	+194

Nomenclature: See separate page